SQL Aggregates and Operators

1) Select ~~Count distinct (department)~~ distinct department as distinct department
From students;

Department
11
HR
Finance

2) Select distinct department, average (age) as avg.age
From students
group by department;

Department	Avg.age
11	20.5
HR	22
Finance	23

3) Select department, count (student.id) as student.count
From students
group by department
having department > 1 student;

Department	Student count
11	2
HR	2

4) Select student.id, name, age, department
From students
where age is between 21 and 23;

student id	name	age	department
2	Bob	22	HR
3	Charlie	21	11
4	Diana	23	Finance
5	Eve	22	HR

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5) Select student_id, name, age, department
from students
where department = 'IT' or 'HR' and age > 21;

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

6) Select department sum(credits) as total credits
from courses
group by department
having sum(credits) as total credits > 5;

department	total credits
IT	11

7) Select *
from students
where courses dont have 4 credits

course_id	course_name	department	cred
101	SQL Basics	IT	3
104	Excel	finance	2
104	statics	HR	3

8) Select course_id, course_name, max(credit) as credits
from courses
limit 3
order DESC,

course_id	course name	credit
102	Python	4
103	Data science	4
101	SQL Basics	3

9) Select max(grade) as max grade, min(grade) as min grade, Avg(grade) as Aggrad
from enrollments

max grade	min grade	Avg grade
90	78	84.6

10) Select course_id, count(*) as enrollment count
from enrollments
Group by course_id;

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

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11) select department, sum(salary) as total salary, sum(bonus) as total bonus
from salaries
group by department;

Department	total salary	total bonus
IT	1222000	10500
HR	109000	7500
finance	70000	6000

12) select department, Avg(salary) as Avg salary
from salaries
Group by department
Having Avg(salary) > 55000;

Department	Avg salary
IT	61000
finance	76000

13) select employee_id, name, salary, bonus, sum(salary + bonus) as total compensation
from salaries
where (salary + bonus) > 60000;

employee id	name	salary	bonus	total compensation
1	Jon	60000	5000	65 000
3	Spike	70000	6000	76 000
4	Tyke	62000	5500	67 5000

14) select department sum(Budget) as total budget, Avg(Budget) as total Avg. budget
from projects
Group by department
having Avg(Budget) > 70000;

Department	total budget	Avg budget
IT	270000	135000
finance	80000	80000

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15) Select * ~~project~~
from projects
where budget 50 000 and 12 0000 and department := 'marketing';

Project Id	Project name	department	Budget
2	payroll system	finance	80000
5	HR Portal	HR	50 000

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